Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Takes logs of an equation. Must be correct use of logs.	AO1.1a	M1	$y = e^{x-4}$ $\ln y = x - 4$ $4 + \ln y = x$ $f^{-1}(x) = 4 + \ln x,\ x > 0$
Obtains correct inverse function in any correct form	AO1.1b	A1	
Deduces correct domain	AO2.2a	B1	
			Total 3 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Integrates using integration by parts	AO3.1a	M1	$y = \int (x-1)e^x \mathrm{d}x$
Applies integration by parts formula correctly to either of $(x - 1)e^x$ or xe^x	AO1.1a	M1	$u = x - 1 \quad \frac{\mathrm{d}v}{\mathrm{d}x} = 1$ $\frac{\mathrm{d}v}{\mathrm{d}x} = e^x \quad v = e^x$
Obtains fully correct integral, condone missing constant.	AO1.1b	A1	$y = (x - 1)e^x - \int e^x \mathrm{d}x$ $y = (x - 1)e^x - e^x + c$
Explains clearly why the minimum y value is e with reference to the range of the function OE	AO2.4	E1	Range ≥ e $\Rightarrow$ at min y = e Min point when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \therefore x = 1$
Uses $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ to find x coordinate of minimum	AO1.1a	M1	So curve passes through (1, e) $e = (1 - 1)e^1 - e^1 + c$
Deduces that the curve passes through the point (1, e)	AO2.2a	A1	$c = 2e$
Uses their minimum point to find their c	AO1.1a	M1	$\therefore f(x) = (x - 2)e^x + 2e$
States the correct equation in any correct form	AO1.1b	A1	

Condone y instead of $f(x)$ CAO			
			Total 8 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Makes a deduction about the lower bound of the function (4)	AO2.2a	B1	
	Correctly states the range of f using set notation	AO2.5	B1	The range of f is the set $\{y \in \mathbb{R} : y > 4\}$
(b) (i)	States correctly the set they gave in part **(a)**	AO1.2	B1F	$\{y \in \mathbb{R} : y > 4\}$
(ii)	Interchanges x and y at any stage	AO1.1a	M1	$y = 4 + 3^{-x}$ $x = 4 + 3^{-y}$
	Rearranges and takes logs	AO1.1a	M1	$3^{-y} = x - 4$ $-y = \log_3(x - 4$
	Obtains correct expression from completely correct working for $f^{-1}(x)$, notation correct throughout	AO1.1b	A1	$f^{-1}(x) = -\log_3(x - 4)$
(c) (i)	Obtains $gf(x)$	AO1.1b	B1	$gf(x) = g(4 + 3^{-x})$ $= 5 - (4 + 3^{-x})^{0.5}$
(ii)	Forms equation and rearranges using 'their' $gf(x) = 2$	AO1.1a	M1	$5 - (4 + 3^{-x})^{0.5} = 2$ $(4 + 3^{-x}) = 9$ $3^{-x} = 5$ $x = -\log_3 5$
	Correctly rearranges to get a single exponential term where logs can be taken. (Follow through provided 'their' equation requires the use of logs.)	AO1.1b	A1F	
	Obtains correct solution	AO1.1b	A1	
				Total 10 marks

Q4.

(a) *f(x)* > − $\frac{4}{3}$, *f* ≥ − $\frac{4}{3}$, *range* ≥ − $\frac{4}{3}$

M1

f(x) ≥ − $\frac{4}{3}$

A1

2

(b) (i) $x \geq -\frac{4}{3}$

correct or FT from (a)

B1F

1

(ii) $x^2 = 3y + 4$

$x = (\pm)\sqrt{3y+4}$

M1

($f^{-1}(x)$ =)(−)$\sqrt{3x+4}$

M1

either order − M1 for correctly changing the subject or reversing operations; M1 for replacing y with x

($f^{-1}(x)$ =)− $\sqrt{3x+4}$

(dependent on both M1 marks) correct sign

A1

3

(c) (i) $3x - 1 = 1$

Or $3x - 1 = e^{\circ}$ or $3x - 1 = \pm 1$

M1

$\frac{2}{3}$ OE

CAO, NMS $\frac{2}{3}$ *OE scores 2/2*

A1

2

(ii) g has **NO** inverse

must indicate no inverse

because two values of x map to one value (of y) **or** it is many-one **or** it is not one-one **or** 'it is two-one'

with valid reason; do not accept contradictory reasons

B1

1

(iii) $\ln\left|3 \times \frac{x^2 - 4}{3} - 1\right|$

M1

$\ln\left|x^2 - 5\right|$

NMS scores 0 / 2, condone k = − 5 after correct expression seen

A1

2

(iv) $\ln\left|x^2 - 5\right| = 0$

$\left|x^2 - 5\right| = 1$

$x^2 - 5 = 1$ (or −1 or e° or −e° seen)

$x^2 - k = 1$ etc, for candidate's positive integer, k

M1

$x^2 = 6$, 4 or candidate's $k + 1$ or $k - 1$

$x = \sqrt{6}$, 2

exact values PI by correct answers

A1F

$x = -\sqrt{6}$, −2

A1F

($x \le 0 \Rightarrow$) $x = -\sqrt{6}$, −2

CAO, rejecting the positive

A1

4

[15]

Q5.

(a) (i) $f(x) = \ln(2x - 3)$

$2x - 3 = e^y$

$2y - 3 = e^x$

Either order:
M1 for antilog
M1 for replacing f(x) or y with x

M1
M1

$(f^{-1}(x) =)\ \frac{1}{2}(e^x + 3)$ OE

Correct expression in x

A1 **3**

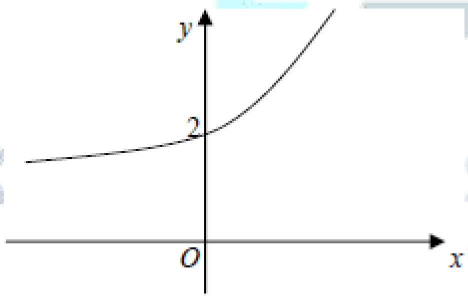
(ii) $f^{-1}(x) > \frac{3}{2}$

*Do **not** condone* $f^{-1}(x) \geq \frac{3}{2}$, $y > \frac{3}{2}$, $x > \frac{3}{2}$

range $> \frac{3}{2}$, $f^{-1} > \frac{3}{2}$

B1 **1**

(iii)



Correct shape crossing y-axis and above x-axis

M1

2 marked on the y-axis

A1 **2**

(b) (i) $(gf(x) =)\ e^{2\ln(2x-3)} - 4$

Correct composition

M1

$= e^{\ln(2x-3)^2} - 4$

PI by correct expression

m1

$= (2x - 3)^2 - 4$

A1

3

(ii) $(\text{fg}(x) =) \ln(2(e^{2x} - 4) - 3)$

OE correct composition

M1

$\ln(2e^{2x} - 11) = \ln 5$

$2e^{2x} - 11 = 5$ OE

Correct antilog of correct equation

A1

$e^{2x} = 8$

$2x = \ln 8$

$x = \frac{1}{2} \ln 8$

OE ***exact solution****, e.g.* $\ln\sqrt{8}$ or $\frac{3}{2}\ln 2$ or $\ln 2^{\frac{3}{2}}$

A1

3

[12]

Q6.

(a)

$\left.\begin{array}{l} f(1) = 21 \\ f(16) = 1 \end{array}\right\}$

sight of 1 and 21

M1

$1 \le f(x) \le 21$

allow f(x) replaced by f, y

A1

2

(b) (i) $y = \frac{63}{4x - 1}$

$x = \dfrac{63}{4y-1}$

M1

$x(4y - 1) = 63$ or better

reverse x, y

one correct step

Either order

M1

$f^{-1}(x) = \dfrac{1}{4}\left(\dfrac{63}{x} + 1\right)$ OE

condone y =

A1

3

(ii) $\dfrac{1}{4}\left(\dfrac{63}{x} + 1\right) = 1$

$\dfrac{63}{x} + 1 = 4$, or better

one correct step from their (b)(i) = 1, or x = f(1)

M1

$(x =)$ 21

note: 21 scores 2 / 2

A1

2

(c) (i) $(fg(x) =) \dfrac{63}{4x^2 - 1}$

B1

1

(ii) $\dfrac{63}{4x^2 - 1} = 1$

$4x^2 - 1 = 63$ or better

one correct step from their (c)(i) = 1

M1

$x^2 = 16$ OE

eg $(2x + 8)(2x - 8) = 0$, or $x = \pm 4$

A1

$x = -4$ **ONLY**

A1

3

[11]

Q7.

(a) $f(x) \geq 0$

$f(x) > 0$, $f \geq 0$, $x \geq 0$, $y > 0$, range ≥ 0

M1

Condone $y \geq 0$

A1

2

(b) (i) $\text{fg}(x) = \sqrt{2\left(\frac{10}{x}\right) - 5}$

$\left(= \sqrt{\frac{20}{x} - 5}\right)$ OE

No ISW

B1

1

(ii) $\sqrt{\frac{20}{x} - 5}$

$\frac{20}{x} = 5^2 + 5$

Correctly squaring their fg(x) and correctly isolating their x term

M1

$x = \frac{2}{3}$

No ISW

A1

2

(c) (i) $y = \sqrt{2x - 5}$

Swap x and y
Correctly squaring } *either order*

M1

M1

$(f^{-1}(x) =)\ \dfrac{x^2+5}{2}$

A1

3

(ii) $x^2 = 9$ or if $\sqrt{9}$ or 3 seen

Candidate must have scored full marks in (c)(i)
(ie no follow through)

M1

$x = 3$ and $x = -3$ rejected

Must see both

A1

2

[10]

Q8.

(a) $-3 \le f(x) \le 3$

$-3 \le x \le 3$, $-3 < f(x) < 3$
$-3 < f < 3$, $-3 < y < 3$
$-3 \le f < 3$, $-3 < f \le 3$

M1

Allow $-3 \le y \le 3$, $-3 \le f \le 3$

A1

2

(b) (i) $y = 3\cos\frac{1}{2}x$

$\frac{y}{3} = \cos\frac{1}{2}x$

$\cos^{-1}\frac{y}{3} = \left(\frac{1}{2}x\right)$

$x = 2\cos^{-1}\frac{y}{3}$

$y = 2\cos^{-1}\frac{x}{3}$

Or $\cos^{-1}\frac{x}{3} =$

Swap x *and* y

Either order

M1

M1

$f^{-1}(x) = 2\cos^{-1}\frac{x}{3}$

A1

3

(ii) $\frac{x}{3} = \cos\frac{1}{2}$

If incorrect in (b)(i) ***BUT*** *answer in form*
$p\cos^{-1}(qx)$ *(condone* $p, q = 1$*)*

Then $qx = \cos\left(\frac{1}{p}\right)$ *M1* ***or*** $x = f(1)$ *M1*

M1

$x = 3\cos\frac{1}{2}$ ISW

$x = 3\cos\frac{1}{2}$ *A1*

A1

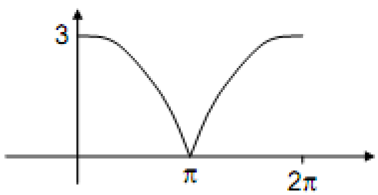
2

(c) (i) $gf(x) = \left|3\cos\frac{1}{2}x\right|$

B1

1

(ii)



Modulus graph in 1st quadrant, starting from
a +ve y*–intercept, at least 2 continuous parts,*

first descending, then second increasing
IGNORE CURVE OUTSIDE RANGE

M1

Correct curvature, curves reaching x–axis, condone multiple curves (no turning points at axis)

A1

Approximately symmetrical graph with 3, π, 2π indicated (must have scored previous 2 marks)

Condone $y = 3\cos\frac{1}{2}x$ *also drawn but clearly identified, otherwise M0*

A1

3

(d) STRETCH + direction

Either in x-direction or y-direction

M1

s.f. 3, parallel to y-axis

A1

s.f. 2, parallel to x-axis

Either order

A1

3

[14]

Q9.

(a) [f(x)] not 1 – 1

OE

E1

1

(b) $y = \dfrac{1}{2x+1}$

$x = \dfrac{1}{2y+1}$

$2y + 1 = \dfrac{1}{x}$

swap x and y
a correct next line $\Big\}$ *either order*

M1

M1

$\left[g^{-1}(x) = \right]\frac{1}{2}\left(\frac{1}{x} - 1\right)$ OE

$[y =]\frac{1}{2}\left(\frac{1}{x} - 1\right)$

A1

3

(c) $[g^{-1}(x)] \neq -0.5$

sight of ≠ –0.5 OE

B1

1

(d) $\left(\frac{1}{2x+1}\right)^2 = \frac{1}{2x+1}$

sight of $\left(\frac{1}{2x+1}\right)^2$ or $\frac{1}{(2x+1)^2}$

B1

$(2x + 1) = (2x + 1)^2$

or $2x + 1 = 4x^2 + 4x + 1$

or $\frac{1}{2x+1} = 1$

or $2x + 1 = 1$

one correct step, must be one of these four lines

M1

$x = 0$

CSO

A1

3

[8]

Q10.

(a) $f(x) > -3$

'> -3', '$x > -3$' or '$f(x) \geq -3$'

M1

Allow $y > -3$

A1

2

(b) (i) $y = e^{2x} - 3$

$y + 3 = e^{2x}$

$\ln(y + 3) = 2x$

swap x and y

M1

attempt to isolate: $\ln(y \pm A) = Bx$ or reverse

M1

$(f^{-1}(x)) = \frac{1}{2}\ln(x + 3)$

OE with no further incorrect working

Condone $y = \ldots\ldots$

A1

Alternative

$x \rightarrow \times 2 \rightarrow e \rightarrow -3$

$\div 2 \leftarrow \ln \leftarrow +3 \leftarrow x$

(M1) (M1)

$y = \frac{\ln(x+3)}{2}$

(A1)

3

(ii) $x + 3 = 1$

for putting their p $(x) = 1$ from $k \ln(p(x))$ in their part (b)(i)

M1

$x = -2$

CSO

SC: B2 $x = -2$ with no working, if full marks gained in part (b)(i)

A1

2

(c) (i) $\left.\begin{array}{l}(\mathrm{g\,f}(x) =) \dfrac{1}{3(\mathrm{e}^{2x} - 3) + 4} \\ (=) \dfrac{1}{3\mathrm{e}^{2x} - 5}\end{array}\right\}$ either OE

substituting f into g
ISW

B1

1

(ii) $\dfrac{1}{3\mathrm{e}^{2x} - 5} = 1$

$1 = 3\mathrm{e}^{2x} - 5$ OE

Correct removal of their fraction

M1

$\mathrm{e}^{2x} = 2$

$2x = \ln 2$

Correct use of logs leading to $kx = \ln \frac{a}{b}$

m1

$x = \frac{1}{2} \ln 2$

CSO No ISW except for numerical evaluation

A1

3

[11]

Q11.

(a) (i) $y = \ln(5x - 2)$

$\dfrac{k}{5x-2}$

M1

$\left(\dfrac{\mathrm{d}y}{\mathrm{d}x} =\right) \dfrac{5}{5x-2}$

No ISW, eg $\dfrac{5}{5x-2} = \dfrac{1}{x-2}$ *(M1A0)*

A1 2

(ii) $y = \sin 2x$

$\left(\frac{dy}{dx} =\right) 2\cos 2x$

A1 2

k cos 2x

M1

(b) (i) $f(x) \geq \ln 0.5$ or $f(x) \geq -\ln 2$

M1A1 2

(ii) $(gf(x) =)$ $\sin [2 \ln (5x - 2)]$
or $(gf(x) =)$ $\sin \ln (5x - 2)^2$

Condone
sin 2 ln (5x – 2) or sin 2 (ln (5x – 2))
*but **not** sin 2 (ln 5x – 2) or sin 2 ln 5x – 2*

B1 1

(iii) $gf(x) = 0$
$\sin[2 \ln(5x - 2)] = 0$
$2 \ln(5x - 2) = 0$

Correct first step from their (b)(ii)

M1

$5x - 2 = 1$

Their f(x) = 1 from k ln (f(x)) = 0

m1

$x = \frac{3}{5}$

Withhold if clear error seen other than omission of brackets

A1 3

(iv) $x = \sin 2y$
$\sin^{-1} x = 2y$ (or $\sin^{-1} y = 2x$)

Correct equation involving $\sin^{-1}$

M1

$(g^{-1}(x) =)\ \frac{1}{2}\ \sin^{-1} x$

A1

2

[12]

Q12.

(a) $f(x) \le 2$, $f \le 2$, $y \le 2$

$\left.\begin{array}{l} \le 2, f(x) < 2, x \le 2 \\ y < 2,\ \ f < 2 \end{array}\right\} B1$

B2

2

(b) $f(x)$ is not one to one

Allow many to one or numerical example

E1

1

(c) (i) $fg(x) = 2 - \left(\frac{1}{x-4}\right)^4$

B1

1

(ii) $2 - \left(\frac{1}{x-4}\right)^4 = -14$

$16 = \left(\frac{1}{x-4}\right)^4$

$\left.\begin{array}{l} (x-4)^4 = \frac{1}{16} \\ x - 4 = \pm\frac{1}{2} \end{array}\right\}$

Correct handling of fourth root
Must have ±

M1

Correct handling of reciprocal

M1

$x = 4\frac{1}{2}, 3\frac{1}{2}$

A1

3

[7]

Q13.

(a) $f(x) \geq 0$

For ≥ 0, f(x) > 0

M1

Correct; allow $y \geq 0$, $f \geq 0$

A1

2

(b) (i) $y = \sqrt{2x + 5}$

$x = \sqrt{2y + 5}$

$x \Leftrightarrow y$

M1

$x^2 = 2y + 5$

Attempt to isolate, squaring first

M1

$f^{-1}(x) = \frac{x^2 - 5}{2}$

condone (y =)

A1

3

(ii) $x \geq 0$

*ft their (a), but **must** be x*

B1F

1

(c) (i) $h(x) = fg(x) = \sqrt{2\left(\frac{1}{4x+1}\right) + 5}$

B1

1

(ii) $\sqrt{2\left(\frac{1}{4x+1}\right)+5}=3$

$2\left(\frac{1}{4x+1}\right)+5=9$

one correct step from (c) (i), squaring

M1

$\left.\begin{array}{l}\frac{1}{4x+1}=2\\ 4x+1=\frac{1}{2}\end{array}\right)$ either or $16x+4=2$

A1

$x=-\frac{1}{8}$ or equiv

CSO

A1 3

[10]

Q14.

(a) all (real) values

No x in answer, unless f(x)

B1 1

(b) (i) fg(x) = $\left(\frac{1}{x-3}\right)^3$

ISW

B1 1

(ii) $\left(\frac{1}{x-3}\right)^3=64$

$\frac{1}{x-3}=4$

$\sqrt[3]{}$

M1

$$x - 3 = \frac{1}{4}$$

Invert

M1

$$x = 3\frac{1}{4}$$

A1

3

(c) (i) $y = \frac{1}{x-3}$

$x = \frac{1}{y-3}$

Swap x and y

M1

$x(y - 3) = 1$

$xy - 3y = 1$

attempt to isolate

M1

$$y = \frac{1+3x}{x} = g^{-1}(x) \text{ or } \frac{1}{x} + 3$$

A1

3

(ii) (real values) $(g^{-1}(x)) \neq 3$

B1

1

[9]

Q15.

(a) $f(x) \geq 0$

allow $f \geq 0$, $y \geq 0$, ≥ 0

B1

1

(b) (i) $y = \frac{1}{2x-3}$

$$x = \frac{1}{2y-3}$$

swap x and y

M1

$x(2y - 3) = 1$
$2xy - 3x = 1$
$2xy = 1 + 3x$

attempt to isolate

M1

$$y = \frac{1+3x}{2x} = g^{-1}(x)$$ o.e.

w.n.f.e

A1

Alternative:
$x \rightarrow$ **×2** $\rightarrow -3 \rightarrow$ **divide into 1** $\rightarrow y$

$$\frac{1}{2y} + \frac{3}{2} \leftarrow \div 2 \leftarrow +3 \leftarrow \text{divide into } 1 \leftarrow y$$

$$\frac{1}{y} + 3$$

M1

3

(ii) $\left(g^{-1}(x)\right) \neq \frac{3}{2}$

B1

1

(c) $$\left(\frac{1}{2x-3}\right)^2 = 9$$

B1

$2x - 3 = \pm\frac{1}{3}$

square root and invert (condone missing ±)
alternative: *attempt to solve a quadratic that comes from*

$$4x^2 - 12 + 9 = \frac{1}{9} \text{ o.e}$$

M1

$x = \frac{5}{3}, \frac{4}{3}$ o.e

A1

3

[8]

Q16.

(a) $f(x) \leq 3$

M1 for f < 3, $x \leq 3$
Condone y, f, range

M1A1

2

(b) (i) $y = \frac{2}{x+1}$

$x + 1 = \frac{2}{y}$

Attempt to obtain x as a function of y or y as a function of x

M1

$x = \frac{2}{y} - 1$

$x \leftrightarrow y$ at any stage

M1

$y / g^{-1}(x) = \frac{2}{x} - 1 = \frac{2-x}{x}$

Any correct form

A1

3

(ii) $(g^{-1}(x)) \neq -1$

B1

1

(c) (i) $h(x) = \dfrac{2}{3 - x^2 + 1}$

M1

$= \dfrac{2}{4 - x^2} = \dfrac{2}{(2-x)(2+x)}$

A1

2

(ii) ($x \in$ **R**), $x \neq +2$, $x \neq -2$

Condone omit 'x is real' Allow $x^2 \neq 4$

B1

1

[9]

Q17.

(a) $f(x) \geq 0$ allow $y \geq 0$

> 0 or $f \geq 0$ or ≥ 0

M1

A1

2

(b) (i) $\sqrt{\dfrac{1}{x} - 2}$

B1

1

(ii) $\dfrac{1}{x} - 2 = 1$

squaring their (b)(i) in an equation

M1

$\dfrac{1}{x} = 3$ OE

A1

$x = \dfrac{1}{3}$

CSO

A1

3

(c) $y = \sqrt{x-2}$

$y^2 = x - 2$

attempt to isolate; condone 1 slip

M1

$x^2 = y - 2$

reverse $x \Leftrightarrow y$

M1

$y = x^2 + 2$

A1

3

[9]

Q18.

(a) (Range of f) ≥ 0

B1

1

(b) (i) fg(x) = $\frac{1}{(x+2)^2}$

OE

Maybe in part (ii)

B1

1

(ii) $\frac{1}{(x+2)^2} = 4$

$(x + 2)^2 = \frac{1}{4}$

Or $4(x + 2)^2 = 1$

M1

$x + 2 = (\pm)\ \frac{1}{2}$

$(2x + 5)(2x + 3) = 0$

M1

$x = -\frac{5}{2}, -\frac{3}{2}$

A1A1

4

(c) (i) Not one to one

OE

E1

1

(ii) $x = \frac{1}{y+2}$

$x \Leftrightarrow y$

M1

$y + 2 = \frac{1}{x}$

Attempt to isolate

M1

$y = \frac{1}{x} - 2$ $\left(\frac{1-2x}{x}\right)$

A1

3

[10]

Q19.

(a) $f(x) = 2e^{3x} - 1$

Range: $f(x) > -1$ (or $y > -1$ or $f > -1$)

for −1 only

M1

exactly correct

A1

2

(b) $y = 2e^{3x} - 1$

$x = 2e^{3y} - 1$

$x \leftrightarrow y$

M1

$2e^{3y} = x + 1$

$e^{3y} = \dfrac{x+1}{2}$

attempt to isolate

M1

$y = \frac{1}{3}\ln\left(\frac{x+1}{2}\right)$

all correct with no error AG (be convinced)

A1

3

(c) $f'^{-1}(x) = \frac{1}{3}\left(\frac{2}{x+1}\right) \times \frac{1}{2}$ OE

for differentiation of ln; $\dfrac{k}{their(x \pm 1)}$

M1

for $\frac{1}{2}$

A1

all correct

A1

$x = 0$

$f'^{-1}(x) = \frac{1}{3}$

CSO

A1

4

Alternative

$f^{-1}(x) = \frac{1}{3}\ln(x+1) - \frac{1}{3}\ln 2$

M1A1

$f'^{-1}(x) = \dfrac{1}{3(x+1)}$

A1

$f'^{-1}(0) = \frac{1}{3}$

CSO

A1

[9]

Q20.

(a) fg = h = $\frac{6}{x+3} - 2$

correct order

M1

$\left(= \frac{6 - 2x - 6}{x+3} = \frac{-2x}{x+3} \right)$

A1

2

(b) (i) $x = \frac{-2y}{y+3}$

$xy + 3x = -2y$

$y(x + 2) = -3x$

attempt to isolate x or y

M1

$h^{-1}(x) = y = \frac{-3x}{(x+2)}$

$\boldsymbol{x} \Leftrightarrow \boldsymbol{y}$

Or:

$y = \frac{6}{x+3} - 2$

$y + 2 = \frac{6}{x+3}$

$x + 3 = \frac{6}{y+2}$

$x = \frac{6}{y+2} - 3$

$h^{-1}(x) = \frac{6}{x+2} - 3$

M1A1

3

(ii) (Range) $\neq -3$

B1

1

[6]